

PRIMEOR SOLUTIONS

E-Commerce Business Performance & Analytics Report

Comprehensive Data Cleaning, SQL Query Mining & Power BI Dashboard Insights

Internship Program:	Data Analytics Internship
Company:	Primeor Solutions
Project Scope:	Levels 1, 2, and 3 Complete Deliverables
Dataset Analyzed:	Global E-Commerce Sales (51,271 Records)
Report Date:	September 2026
Submission Format:	Packaged Final ZIP Archive

Executive Summary: This technical business report synthesizes the end-to-end data analytics workflow completed for Primeor Solutions. Starting from raw e-commerce transaction logs tainted with missing customer names, duplicate rows, and conflicting date formats, we constructed an automated cleaning pipeline, loaded the refined relational schema into SQL, executed advanced queries to identify profit centers and loss leakages, and designed an executive multi-page Power BI dashboard.

1. Level 1: Data Understanding & Cleaning Pipeline

The raw e-commerce dataset comprised **51,325 transactions** across 21 attributes. Prior to exploratory data analysis, systematic data hygiene operations were conducted to guarantee data integrity, eliminate anomalies, and conform to relational modeling standards.

Cleaning Operation	Issue Identified in Raw Data	Resolution & Methodology
Deduplication	54 exact duplicate rows detected across all attributes.	Removed duplicates, reducing volume to 51,271 distinct rows.
Missing Customer Names	79 missing entries in customer_name column.	59 rows mapped from existing order_id relations; 20 imputed as 'Unknown Customer'.
Missing Profit Values	14 null entries in profit field.	Imputed using sub-category median profit-to-sales ratios * sales.
Mixed Date Formats	Inconsistent date notations (20,087 with '/' vs 31,238 with '-').	Standardized 100% of order_date and ship_date to ISO 8601 (YYYY-MM-DD).
Numeric Formatting	Sales figures stored as strings with thousand-separator commas.	Sanitized strings and converted to standard float64, rounded to 2 decimals.
Text Inconsistencies	Leading/trailing whitespace and inconsistent casing.	Stripped whitespace across all categorical dimensions.

Excel Analysis Highlights (Task 2 & 3)

- **Global Gross Sales: \$12,642,501.91** across 51,271 order line items.
- **Global Net Profit: \$1,474,183.04**, delivering a robust **11.66% net margin**.
- **Average Global Discount: 14.29%** (with discounts reaching up to 85% in select regions).
- **Top Product by Revenue:** *Apple Smart Phone, Full Size* (\$86,935.78 in gross sales).
- **Top Loss-Making Product:** *Cubify CubeX 3D Printer Double Head* (-\$8,879.97 net loss).

2. Level 2: SQL Analytics & Relational Mining

The cleaned dataset was ingested into an SQLite database (primeor_ecommerce.db) within table sales_data. Below is a synthesized breakdown of the core SQL queries and findings.

Market-Wise Revenue and Profitability (Query 8)

Market	Orders	Revenue (\$)	Profit (\$)	Margin %
APAC	10,997	\$3,581,785.00	\$437,044.95	12.20%
EU	9,998	\$2,937,166.00	\$372,884.62	12.70%
US	9,990	\$2,297,022.00	\$286,191.42	12.46%
LATAM	10,291	\$2,163,919.00	\$221,299.54	10.23%
EMEA	5,026	\$805,053.00	\$43,810.19	5.44%
Africa	4,585	\$783,345.00	\$88,848.17	11.34%
Canada	384	\$66,932.00	\$17,821.82	26.63%

Category Performance & Margin Analysis (Queries 4 & 5)

Category	Transactions	Total Sales (\$)	Total Profit (\$)	Avg Discount	Margin %
Technology	10,137	\$4,742,975.00	\$663,340.48	13.54%	13.99%
Office Supplies	31,267	\$3,786,476.00	\$518,376.57	13.74%	13.69%
Furniture	9,867	\$4,105,771.00	\$286,183.66	16.80%	6.97%

Shipping Tier Utilization & Logistics Costs (Query 10)

Shipping Mode	Total Orders	Volume Share %	Total Sales (\$)	Avg Shipping Cost (\$)
Standard Class	30,764	60.00%	\$7,574,395.00	\$19.96
Second Class	10,304	20.10%	\$2,563,078.00	\$30.44
First Class	7,504	14.64%	\$1,830,652.00	\$41.04
Same Day	2,699	5.26%	\$674,377.00	\$86.91

3. Executive Answers to Core Project Questions

Q1: Which market generates the highest revenue?

- **Primary Driver: APAC (Asia-Pacific)** is the dominant global revenue engine, generating **\$3,581,785.00** (28.3% of total revenue) with **\$437,044.95** in net profit.
- **Growth Factors:** Rapid adoption of high-ticket telecommunications hardware (Smart Phones) and imaging copiers across Australia, China, and India.
- **Secondary Markets:** Europe (EU) ranks second (\$2.94M revenue, 12.7% margin), followed closely by the United States (\$2.30M revenue, 12.5% margin).

Q2: Which categories are least profitable?

- **Lowest Profit Category: Furniture** is the least profitable category, capturing only **6.97% net margin** despite contributing over \$4.10M in sales.
- **Sub-Category Loss Leader:** The **Tables** sub-category generated an aggregate net loss of **-\$64,113.55** (profit margin: **-8.48%**).
- **Root Cause:** Furniture suffers from the highest average discount rate (**16.80%**, reaching up to 85% during promotional campaigns) combined with elevated freight shipping expenses for bulky items.

Q3: Which shipping mode is most commonly used?

- **Dominant Tier: Standard Class** accounts for exactly **60.00% of all customer orders** (30,764 orders).
- **Cost Efficiency:** Average shipping cost for Standard Class is **\$19.96**, compared to \$86.91 for Same Day delivery. Customers strongly prioritize free/discounted standard shipping over speed.

Negative Profit Orders Analysis (Leakage Mitigation)

Our SQL analysis identified **12,540 order transactions** that yielded negative profit margins, totaling over **\$500,000 in erased gross profit**. Key observations:

1. Over 88% of loss-making transactions featured discounts exceeding **30%**.
2. High-value tech equipment (e.g. 3D Printers) discounted at 50-70% generated losses up to **-\$6,599.98 per order**.
3. Sean Miller generated \$35,131 in revenue but recorded a net **loss of -\$397.21** for the company due to excessive bulk purchase discounting.

Strategic Business Recommendations

- 1. Implement Strict Discounting Guardrails:** Enforce a hard discount ceiling of 20% on Furniture and 15% on high-ticket Technology. Require VP-level sign-off for discounts > 25% to immediately halt margin erosion.
- 2. Restructure the Tables Sub-Category:** Phase out unprofitable table SKUs or re-negotiate supplier pricing. Introduce mandatory customer-paid bulky shipping surcharges for Furniture orders.
- 3. Turnaround Plan for EMEA & Southeast Asia:** Southeast Asia delivers only a 2.00% profit margin and EMEA delivers 5.44%. Rationalize localized pricing models, eliminate deep promo discounting, and focus on high-margin Technology products in these territories.
- 4. Loyalty Rewards on Standard Shipping:** Since 60% of customers choose Standard Class, bundle shipping discounts into a subscription/loyalty model (e.g. Primeor Plus) to drive repeat purchase frequency while stabilizing fulfillment costs.

4. Level 3: Power BI Dashboard Architecture

The Power BI analytics suite is designed across three cohesive report pages adhering to institutional design standards with a modern palette (Navy #1B365D, Teal #00A896, Slate #4A5568, Gold #F5A623):

Report Page	Target Persona & Scope	Core Visuals & KPI Cards Included
Page 1: Executive Overview	C-Suite & Commercial Leadership	• 4 KPI Cards: Total Sales, Net Profit, Order Volume, Avg Discount • Clustered Column: Category Sales vs. Profit • Donut Chart: Market Revenue Share
Page 2: Sales Analysis	Regional VP & Sales Directors	• Bar Chart: Top 8 Geographic Regions • Column Chart: Global Market Breakdown • Line/Area Chart: 48-Month Revenue Trend (Seasonality) • Column Chart: Profit Margin % by Category
Page 3: Product & Customer Intelligence	Product Managers & Merchandising	• Horizontal Bar: Top 5 Profitable Products • Horizontal Bar: Top 5 High-Value Customers • Tornado Bar: Net Profit by Sub-Category (Loss Flagging) • Column Chart: Sales & Profit by Customer Segment

Interactive Slicers & Dynamic Filtering

The dashboard incorporates cross-filtering slicers for **Order Year (2011 - 2014)**, **Market**, **Region**, and **Customer Segment**. Stakeholders can isolate specific seasonal campaigns, regional territories, or customer cohorts with instantaneous visual recalculation.

Deliverables Summary & Package Architecture

All project assets have been generated, validated, and organized into dedicated level directories as mandated by Primeor Solutions guidelines:

Level / Module	Generated Deliverable Files	Status
Level 1: Excel	• Cleaned_Dataset.csv • Cleaned_Dataset.xlsx • Excel_Analysis_Workbook.xlsx (Formulas, KPIs, 4 Pivot Charts)	100% Complete
Level 2: SQL	• primeor_ecommerce.db (SQLite database) • sql_queries.sql (10 Commented Production Queries) • sql_insights_summary.md	100% Complete
Level 3: Power BI	• PowerBI_Measures_and_Guide.md (DAX Formulas) • interactive_dashboard.html (Standalone Interactive App) • page1, page2, page3 high-res dashboard renders (.png)	100% Complete
Final Submission	• Final_Business_Insights_Report.pdf • Primeor_Solutions_Internship_Final_Submission.zip	100% Complete